First and Second Order Modling of DC motor

2.1 Aim

Transient performance analysis of DC Motor speed control circuit and Identify the First and Second order transfer function.

2.2 List of Equipment

Table 2.1: List of Equipment

S.No.	Equipment Name	Manufacturer
1	Encoder Motor	1
2	STM32 Microcontroller	F767zi
3	Breadboard	1
4	Motor Driver	1

2.3 Theory

The transient performance of a DC motor speed control circuit can be analyzed by examining its response to a step input. The response of the system can be characterized by its time constant, rise time, settling time, steady-state eror and overshoot. A first-order transfer function is given by:

$$G(s) = \frac{K}{\tau s + 1} \tag{2.1}$$

where K is the system gain and τ is the time constant.

A second-order transfer function is given by:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{2.2}$$

$$\text{or} \quad G(s) = \frac{K}{as^2 + bs + c} \tag{2.3}$$

where ω_n is the natural frequency and ζ is the damping ratio.

The time constant τ of a first-order system is the time it takes for the system to reach approximately 63.2% of its final value. For a second-order system, the damping ratio ζ determines the nature of the transient response. If $\zeta < 1$, the system is underdamped and exhibits oscillatory behavior. If $\zeta = 1$, the system is critically damped and returns to equilibrium without oscillating. If $\zeta > 1$, the system is overdamped and returns to equilibrium slowly without oscillating.

By analyzing the step response of the DC motor speed control circuit, we can identify the parameters of the transfer function and determine whether the system behaves as a first-order or second-order system.

2.4 Procedure

2.4.1 Design and Speed measurement of Dc motor via Encoder

1. Connect the DC motor speed control circuit as per the schematic diagram figure 2.1.

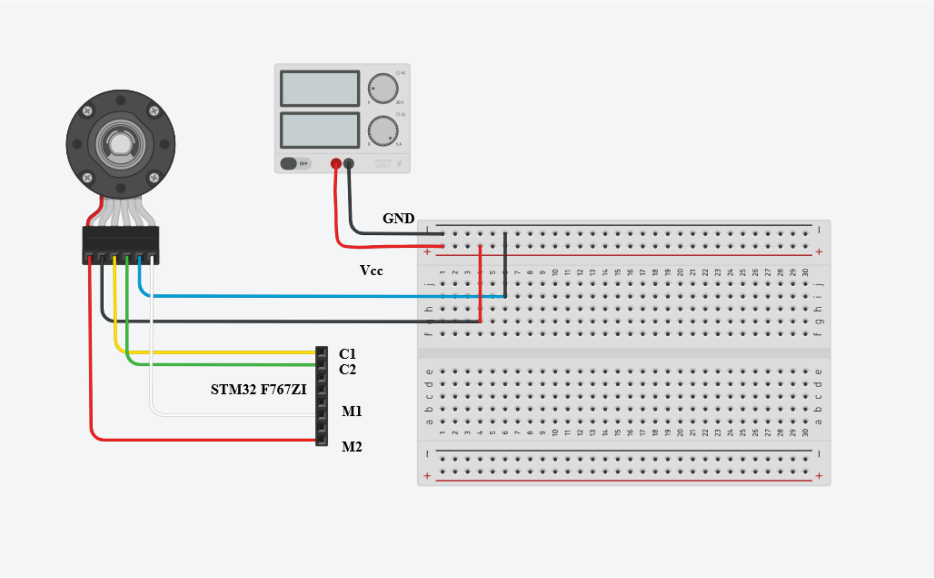

Figure 2.1: Circuit Diagram.

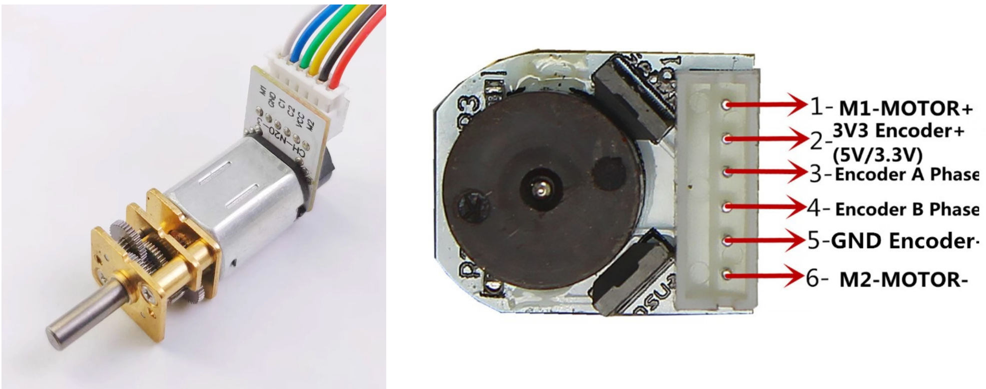

Figure 2.2: Encoder Pin Diagram

2. Apply a step input duty cycle to the motor to simulate motor at various speed.
3. The speed various is achieved by Pulse width modulation (PWM) on specified pins and record the speed response as shown in figure 2.4.
4. To measure the time period and simulate the low pass filter transfer function at the frequency obtained by the measured time period.

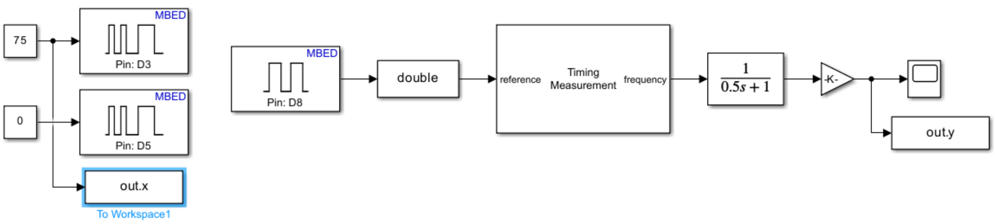

Figure 2.3: Motor Speed Response under Step input Duty cycle

5. Measure the time taken by the motor to reach its steady-state speed.
6. Save the input and output data to workspace using “To workspace” block. Set block format to “Timeseries as output as shown in figure no. 2.5.

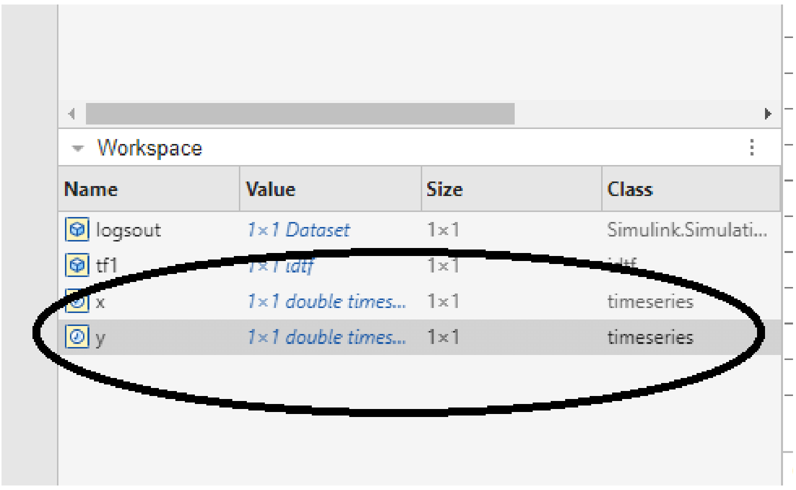

Figure 2.4: Stored data in workspace.

7. Calculate the time constant and other parameters based on the observed data.

$$\tau = t_{63.2\%} - t_0$$

$$K = \frac{\omega_{final}}{\omega_{input}}$$

where $t_{63.2\%}$ is the time at which the speed reaches 63.2% of its final value, and t_0 is the initial time when the input was applied.

8. Repeat the experiment for different input voltages to analyze the system's behavior under varying conditions.

9. Plot the speed response against time for each input voltage.

10. Analyze the plots to identify the second-order characteristics of the system.

2.4.2 Second Order System Identification

The second-order system model can be identified by using the system identification toolbox in MATLAB. The steps are as follows:

1. Repeat step no 1 to 4.

2. Open "System Identification Toolbox" in MATLAB.

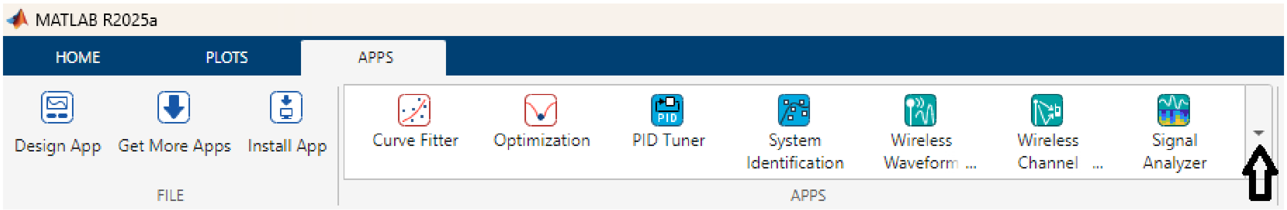

Figure 2.5: Step 1: Open Apps in MATLAB.

3. Launch system Identification app

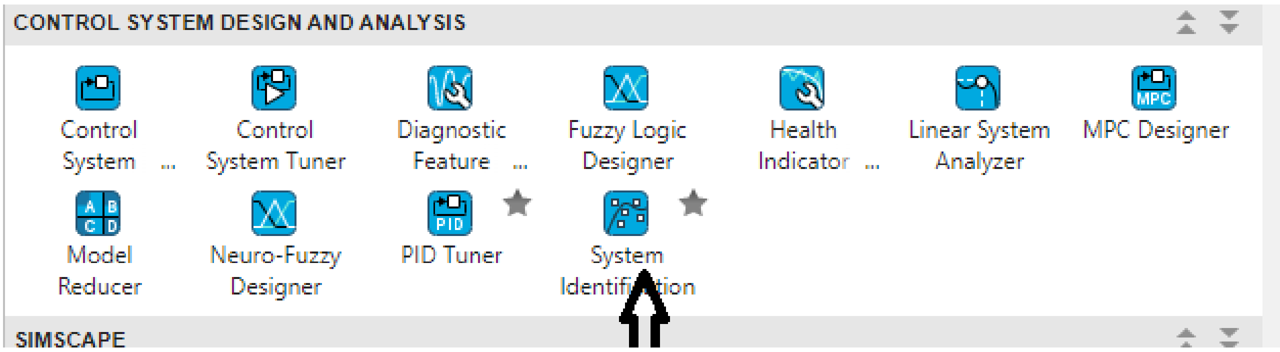

Figure 2.6: Step 2: Launch System Identification app.

4. Import and prepare data.

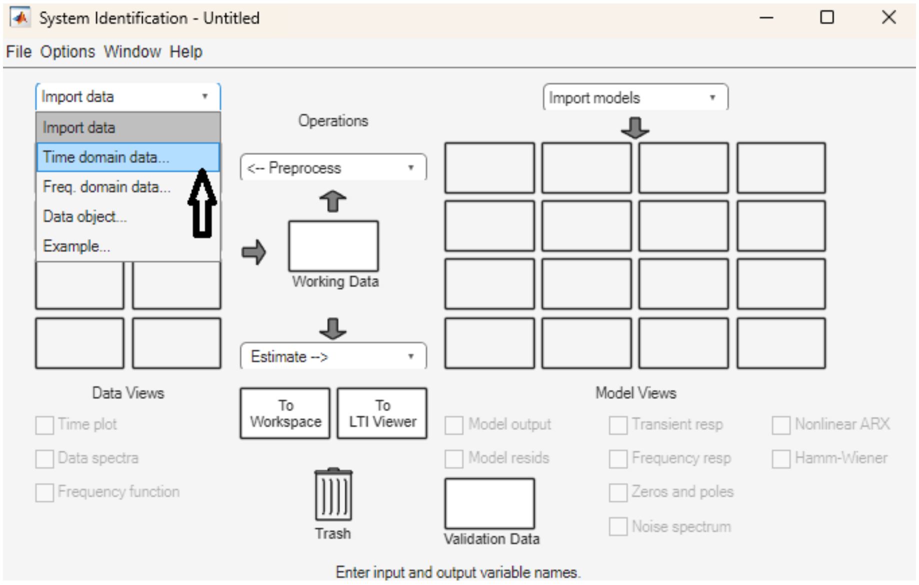

Figure 2.7: Step 3: Import Data → Time-domain data.

5. Select input-output data from the workspace then press Import.

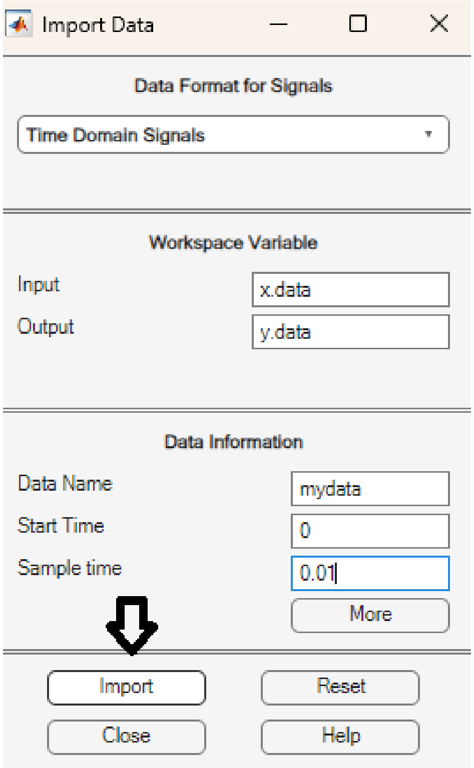

Figure 2.8: Step 4: Select input-output data.

6. Choose model type for identification.

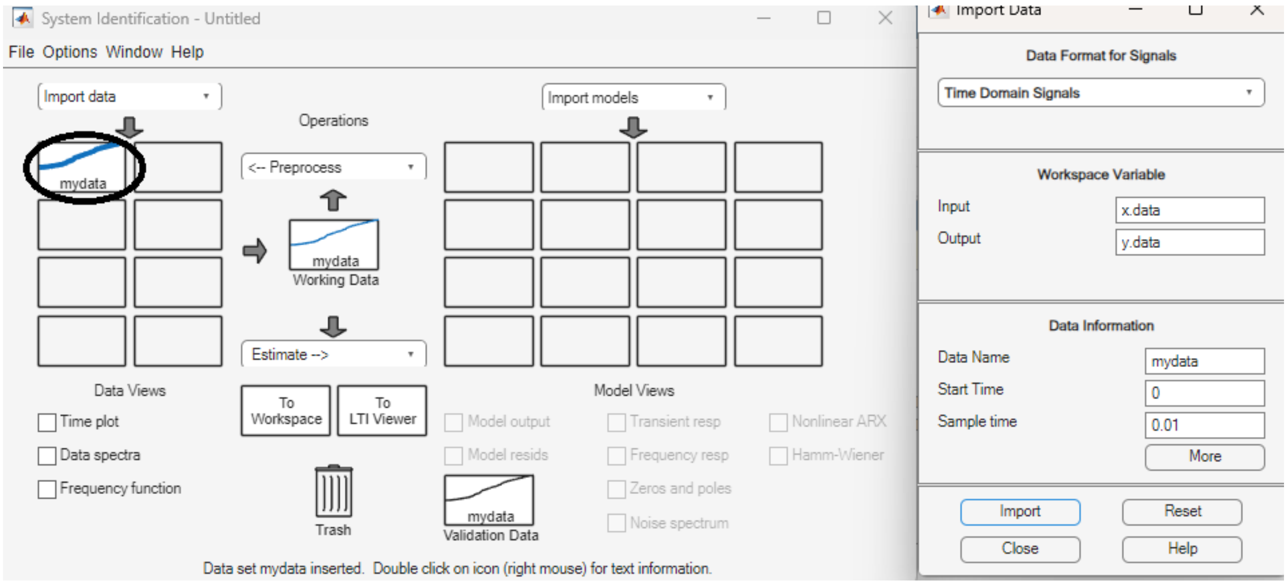

Figure 2.9: Step 5: Choose model type for identification.

7. To estimate the model parameters, select the transfer function models then press Estimate.

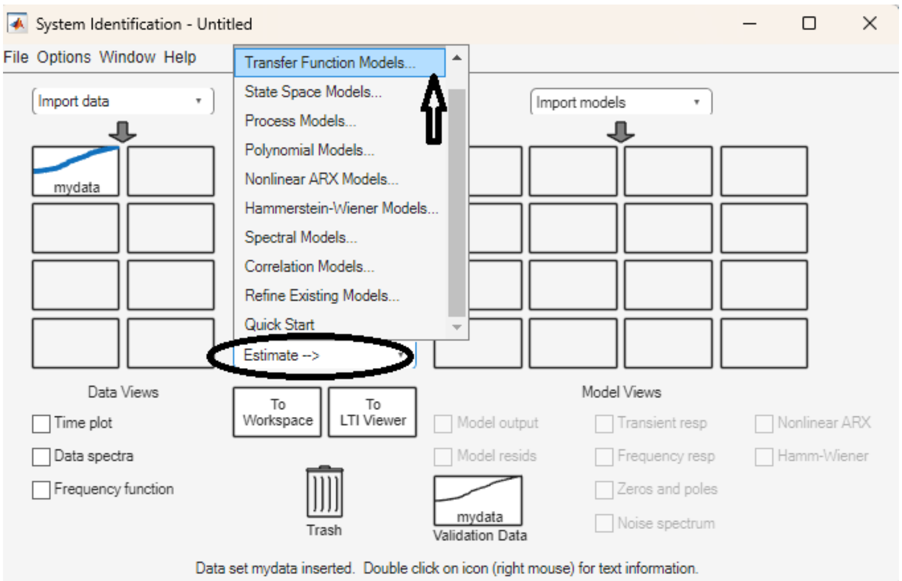

Figure 2.10: Step 6: Estimate model parameters.

8. For the second-order model, select the number of poles and zeros, and then perform the estimate.

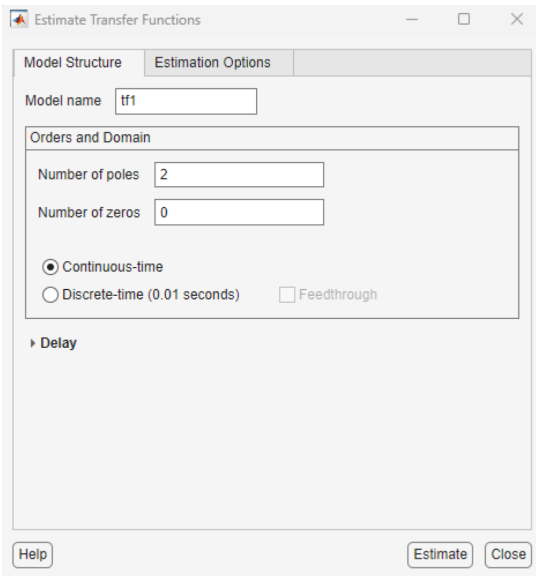

Figure 2.11: Step 7: Compare model with measured data.

9. Analyze the system and check the model accuracy in percentage.

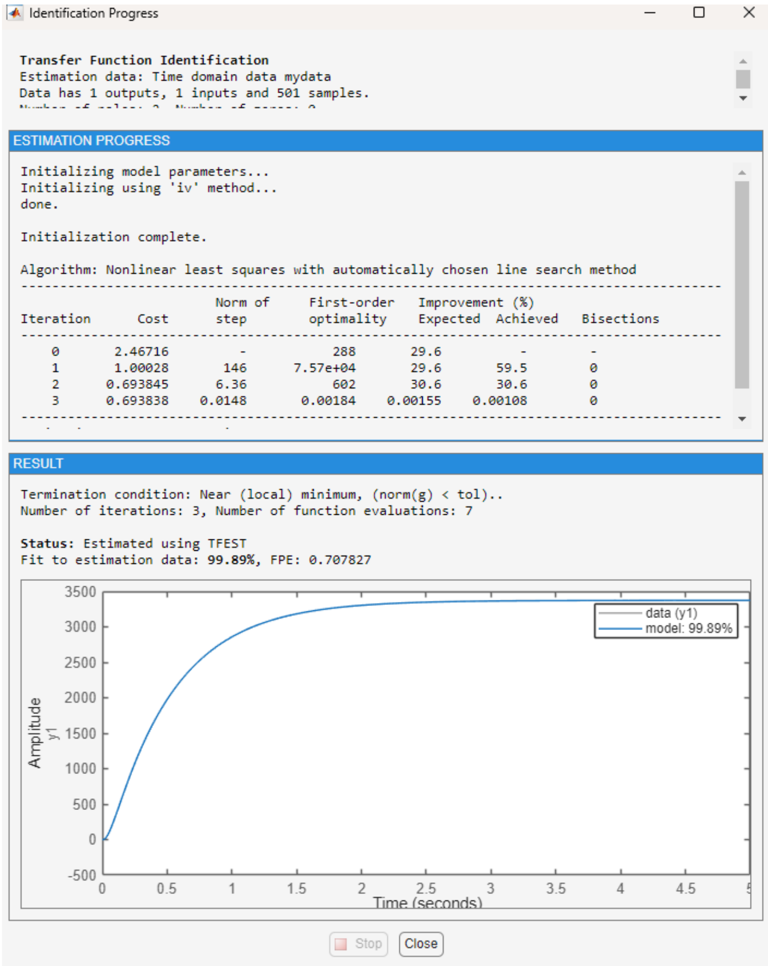

Figure 2.12: Step 8: Analyze the system.

10. Right-click on tf1 to obtain the transfer function.

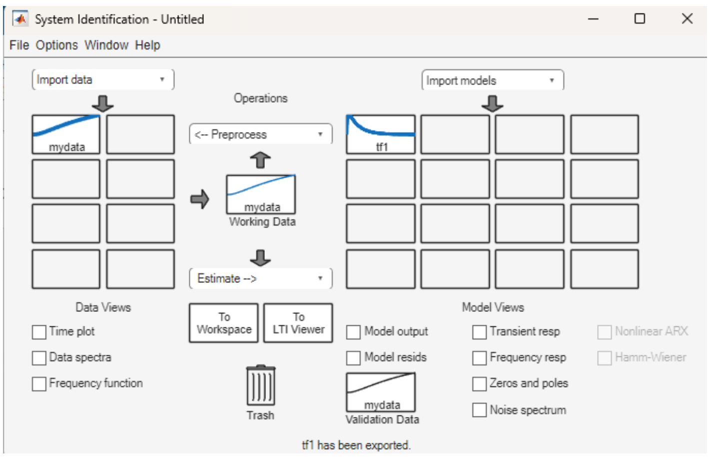

Figure 2.13: Step 9: Transfer Function of DC motor.

11. Export identified model to get the data into the workspace.

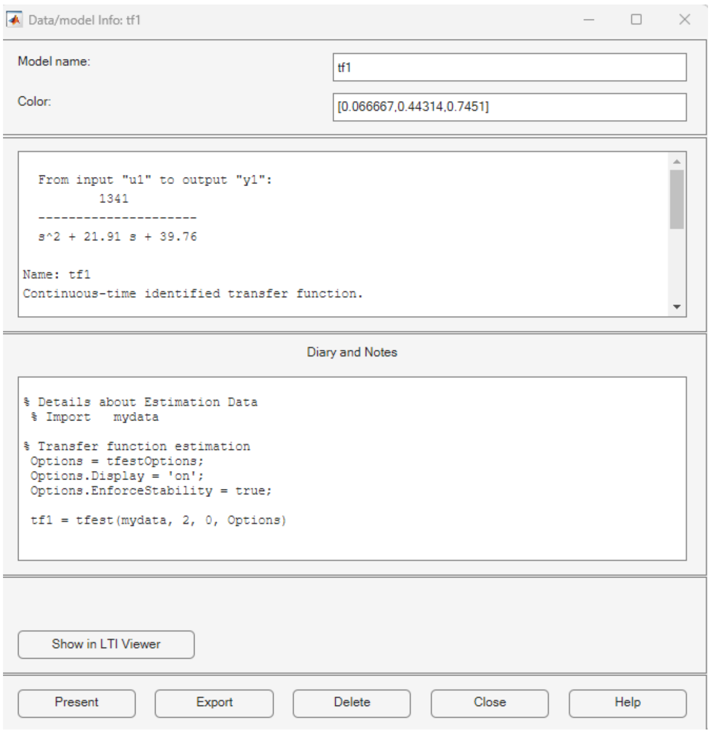

Figure 2.14: Step 10: Export model to MATLAB workspace.

12. Export identified model to get the data into the workspace.

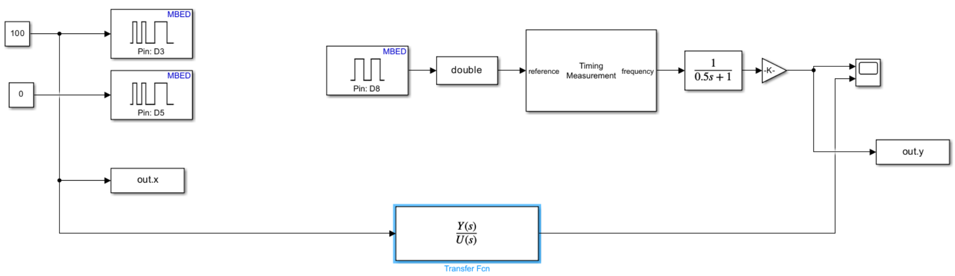

Figure 2.15: Simulink Model for Hardware Implementation.

13. Find the data (input and output) in the MATLAB workspace.

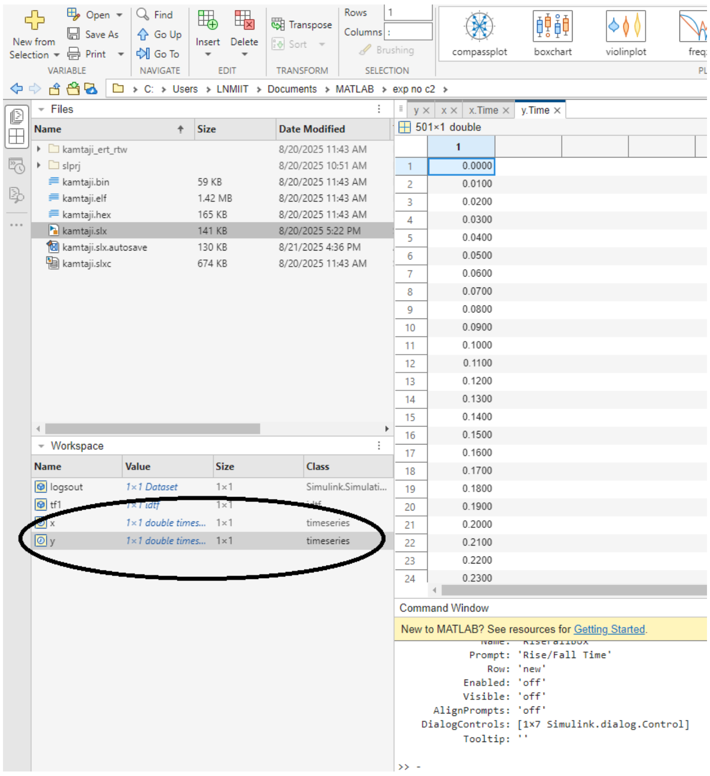

Figure 2.16: Step 11: Input Output data in the MATLAB workspace.

14. Load the transfer function data into MATLAB Simulink.

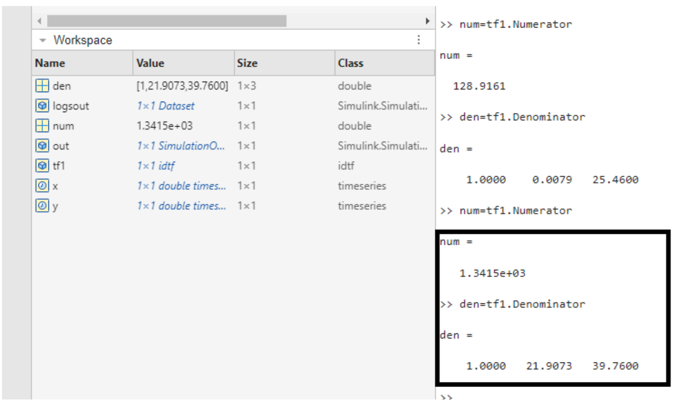

Figure 2.17: Step 11: To get Numerator and denominator data.

15. Compare the transient response of the DC Motor and its second-order transfer.

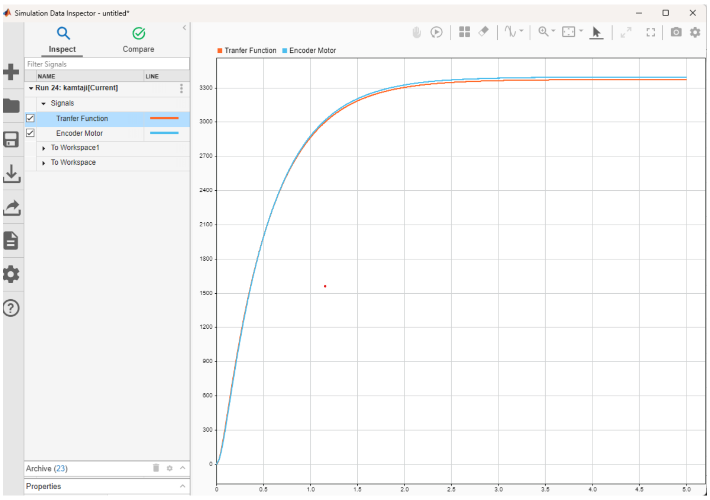

Figure 2.18: Step 11: Comparison of Step response.

2.5 Observation

2.5.1 First Order System

The system is modeling into the following transfer function:

$$G(s) = \frac{K}{\tau s + 1}$$

Table 2.2: Observation Table for First Order System

S.No.	Duty Cycle (D)	Speed (RPM)	K	τ	ISE
1	20%	70			
2	40%	100			
3	50%	150			
4	75%	175			
5	100%	250			

2.5.2 Second Order System

The system is modeled into the following transfer function:

$$G(s) = \frac{K}{as^2 + bs + c}$$

where K is the system gain, and a, b, and c are constants that define the system dynamics.

Table 2.3: Observation Table for Second Order System

S.No.	Duty Cycle (D)	Speed (RPM)	K	a	b	c	ISE
1	20%	70					
2	40%	100					
3	50%	150					
4	75%	175					
5	100%	250					

2.6 Result and Conclusion

The step response of the DC motor speed control circuit was analyzed, and the system was identified as a second-order system. The transfer function parameters were determined using the system identification toolbox in MATLAB. The identified model accurately represented the transient response of the DC motor, confirming its second-order characteristics. The analysis provided insights into the system's dynamics, which can be used for further control system design and optimization.